

PTC Uniform/Constant Analysis Coefficient

Engineering Conformance Specification

SCI-PTC-COEFFICIENT-UNIFORM v0.1-draft.1 / r0.2

Implementation-blind scientific author

2026-09-05

Abstract

Draft for scientific-owner review. This view renders the same canonical definitions, equations, assumptions, requirements, and edge cases as the scientist-facing contract. Requirement verbs describe proposed eventual conformance obligations and do not authorize implementation or claim that any test has passed.

1 Program adherence and prior-work recovery

This implementation-blind first scientific draft follows the released Scope Brief, its completed prior-work recovery synthesis, and the admitted program charter excerpts. It adopts the frozen PTC coefficient taxonomy, generic coefficient interface, immutable lifecycle, named-use separation, and PTC-disabled route; the approved single-registry framework; the frozen MAP and JINC coefficient-boundary constraints; the frozen NOI balance and lossless rational-parent obligations; and the admitted VAL knowledge, binding, and four-axis rules. It cites the generic PTC construction instead of re-deriving it. Implementation, tests, audits, repairs, validation, and external scientific material were excluded.

The genuinely new content is one proposed uniform/constant family and one complete proposed PTC-owned coefficient-handoff profile. The family, profile, registry key, two boundary successors, numerical definition, normalization, support, permissions, and policies are proposals awaiting scientific-owner review. No request default or mode policy is proposed. Sensitivity, residual-scatter, hybrid, learned, and adaptive families remain deferred.

Compact input/output/equation/source/status block

Input	One exact realized PTC transformed product Z ; its immutable CAL parent; application generation; observation and occurrence identities; output support; requested/effective/resolved/realized coefficient selection; and preserved typed facts.
Output	A proposed PTC-owned positive analysis/gridding coefficient ω_i (also denoted γ_i at the MAP boundary), its immutable family generation and payload, and a complete named-use decision record per requested occurrence and named consumer.
Equation	$\tilde{\omega}_i = 1$ and $\omega_i = \tilde{\omega}_i / (\mathcal{D} ^{-1} \sum_{j \in \mathcal{D}} \tilde{\omega}_j) = 1$ for every $i \in \mathcal{D}$, with $\mathcal{D}$ exact and nonempty.
Sources	Released charter/scope/recovery material; frozen PTC, MAP, JINC, and NOI excerpts; approved PTC registry framework; admitted VAL Core, Registry, and PTC common-fragment excerpts.
Status	Algebraically complete author proposal. Owner adoption, implementation conformity, representation fidelity, observational performance, and production readiness are not established.

Symbols

o is one observation; p is one exact realized PTC transformed-product and application generation; g is one immutable coefficient generation bound to p ; i is one exact detector-time occurrence; ι_i is its complete stable identity; $\mathcal{S}_{p,o}^{\text{PTC,out}}$ is the exact PTC output support; $\mathcal{D}_{g,o}$ is the coefficient normalization population; $\tilde{\omega}_i$ and ω_i are the raw and normalized PTC coefficients; γ_i is MAP's notation for the same PTC value; and κ_{ip} is the separately JINC-owned signed spatial factor.

2 Physical meaning and intended use

The family assigns equal PTC coefficient influence to every occurrence in one explicitly declared coefficient support. It is a transparent baseline: relative differences in downstream contribution cannot originate from this PTC coefficient. Other producer and consumer factors can still make total contributions unequal. MAP retains its own admission, placement, boundary, exposure, and contribution rules. JINC retains its signed spatial factor κ_{ip} and its own normalization, conditioning, and support.

Uniformity is a statement only about the PTC-owned positive coefficient. It does not assert equal signal quality, equal exposure, equal response, equal uncertainty, detector independence, inverse variance, sensitivity, significance, precision, or optimality. It does not restore an invalid sample, create output support, or override any consumer decision.

3 Common registry-entry contract

Facet	Proposed exact content
Registry/family identity	SCI-PTC:analysis_gridding_coefficients@draft-0.1; family SCI-PTC:uniform_constant@draft-0.1; scientific owner SCI-PTC. The proposed actual human owner binding is Grant Wilson acting for SCI-PTC; exact confirmation is an open owner decision. Both identities remain drafts until owner adoption and immutable Registry binding.
Index and broadcast	Logical detector-time occurrence i with full exact identity. One scalar may represent the family only through the lossless target broadcast in Equation 1; a complete stored membership manifest is separately checked against its bound member identities.
Statistic and factors	No measured statistic and no fitted, residual, sensitivity, exposure, response, covariance, or loading factor. The sole raw factor is fixed $c_\star = 1$.
Unit	Dimensionless unity. The unit conveys no precision or uncertainty meaning.
Normalization	Named operator $\mathcal{N}_{\text{mean1}}$ in Equations 5–6; exact population $\mathcal{D}_{g,o}$ in Equation 2.
Estimation population	None: the value is defined rather than estimated. The normalization population still identifies its exact scientific scope.
Support	Exactly the bound product's PTC output support for one observation. Fit and application supports remain separate producer facts.
Parent compatibility	Exact PTC transformed product/application generation, immutable CAL parent, observation, occurrence identities, and selection lineage must match. Numerical equality establishes no compatibility.
Uncertainty meaning	The coefficient has no sampling uncertainty because it is an exact policy constant. This supplies no zero for signal, calibration, response, covariance, selection, representation, or performance uncertainty; those states remain separate.
Lifecycle	Requested, effective-policy, observation-resolved, selected, realized, published, and evaluation states are immutable and one-way linked. A re-realization creates a new g and never modifies p or an earlier g .
Consumer permissions	Separate proposals to permit SCI-MAP and SCI-JINC. Each requires owner adoption. Neither can be inferred from the other.
Numerical use	Positive multiplicative analysis/gridding coefficient in already-governed MAP or JINC consumer arithmetic.
Prohibited meanings	Precision, inverse variance, sensitivity, significance, fitted loading, residual scatter, validity, exposure, response, covariance, independence, optimality, default, support restoration, or missing-value substitution.
Missing/invalid/conflict	Missing selection, family, structural binding, support, payload/fidelity evidence, profile decision, or permission makes the affected route unavailable. A complete authoritative false makes the payload producer-invalid and ineligible. Structural conflict stops the question. No state becomes zero or unity by fallback.
Unrepresentable state	After structural establishment, an authoritative complete finding that the bound representation cannot encode exact dimensionless one or its exactly-once stored membership is F and producer-invalid. Missing representation/decoder evidence is U . A negative profile evaluation may itself be successfully realized.

4 Proposed complete coefficient-handoff profile

4.1 Identity, use, and object

The proposed profile is `SCI-PTC:uniform_coefficient_handoff@draft-0.1`. Its scientific package/producer owner is SCI-PTC, and its proposed actual-owner field is Grant Wilson acting for SCI-PTC, pending exact confirmation. Its exact use is:

For one requested tuple (g, o, i, c) , decide whether a realized PTC coefficient payload is a faithful instance of `SCI-PTC:uniform_constant@draft-0.1` and is authorized for handoff to the explicitly named consumer $c \in \{\text{SCI-MAP}, \text{SCI-JINC}\}$.

The object is one coefficient occurrence plus its generation-level scalar and broadcast binding. The profile binds the admitted PTC common named-use fragment and the exact eventual owner-approved source bytes of this contract. It is not the unsupported generic name `SCI-PTC:coefficient_qc_population@1`; it is a complete narrower proposition. While its source or Registry record remains draft or absent, binding integrity is not authoritative and no eligible evaluation may be claimed.

4.2 Applicability and structural gate

The profile is requested only when the exact family is the resolved selection for a named consumer and a coefficient handoff is requested. It is applicable when the exact PTC product exists, $\mathcal{D}_{g,o}$ is authoritatively nonempty, and $i \in \mathcal{D}_{g,o}$. It is inapplicable when authoritative identity establishes $i \notin \mathcal{D}_{g,o}$. It has `applicability_unknown` when product, observation, support, target occurrence membership, or generation binding is missing or conflicting.

The nonexceptionable structural gate requires an active exact Registry record; actual owner and source binding; complete occurrence, observation, product, parent, family, generation, selection-lineage, support, payload-publication, target-broadcast, profile, and named-consumer identities; and consistent scopes and immutable provenance. The target population must contain exact unique occurrence identities, and the target broadcast must bind the identified publication slot to that population and generation. An expected publication slot may be identified while its bytes remain absent or unavailable; that availability is a later payload-fidelity input. Missing or conflicting target broadcast binding, duplicate or ambiguous occurrence identity, inferred ordering, cross-observation pooling, or cross-generation pairing fails the gate and makes the decision unavailable rather than use-specifically ineligible.

Structural establishment fixes what population and publication are being asked about; it does not assert that the stored payload faithfully realizes them. Only after that question exists does UQ-05 compare the bound bytes and a complete stored payload-membership manifest with the authoritative target. Thus a duplicate identity in the authoritative population is structural, whereas a repeated stored payload member relative to an already unique target population is a payload-fidelity violation.

4.3 Exact profile inputs and truth-domain rules

The profile consumes only the following owner-bound inputs after the structural gate establishes their identities: the resolved family-selection record and its parameter field; the PTC compatibility permission for the already-bound product and coefficient generations; the immutable PTC output-support record and exact member identities; the family-definition record naming constant, unit, operator, and population; the typed payload-availability record, published scalar bytes, declared decoder, decoded unit, complete stored payload-membership manifest, publication digest, and authoritative integrity result; the exact named-consumer permission; and immutable lifecycle links and overwrite/nonmutation attestations. Sample validity, retention, consumer admission, response, covariance, and diagnostics are preserved inputs for provenance but are not permission predicates.

For each restriction below, **T** means its complete authoritative input set positively satisfies the stated predicate; **F** means the complete authoritative input set explicitly violates it; **U** means a required input or complete assertion is missing or the check was not performed; and **C** means authoritative predicate inputs for the same structurally established object are contradictory or ambiguous. A parent, occurrence, source, or generation identity mismatch is handled earlier by the structural gate and is therefore decision unavailable. It is not converted to **F** here.

The payload-fidelity input in UQ-05 is the producer QC primitive, not an assertion of this profile's outcome. It is evaluated only after the exact unique $\mathcal{D}_{g,o}$, payload-publication identity, and target broadcast binding are structurally established. Its disjoint truth-domain rule is exact:

- **T** only when SCI-PTC authoritatively reports `computed_published`, the bound bytes are readable under the declared decoder, the decoded scalar and unit are exactly (1,1), the publication digest matches, and a complete authoritative stored membership manifest contains every and only each target identity exactly once;
- **F** only when a complete authoritative fidelity check establishes a decoded non-one value or wrong unit; a digest or other integrity failure; corruption, truncation, decoder-invalid bytes, or known unrepresentability, including a corrupt scalar that cannot be read; or a complete stored membership manifest that omits, adds, or repeats any member relative to the established target population;
- **U** when the payload, bytes, decoder, digest, integrity result, complete check, or complete stored membership manifest is absent or unavailable, or when the payload is unreadable without an authoritative fidelity cause or verdict; and
- **C** when authoritative fidelity records for the same bound payload contradict one another or are ambiguous. The conflict is preserved and composes as unresolved **U** unless another required predicate is decisively **F**.

Missing or conflicting structural identity never enters this UQ-05 partition; it stops the question at the structural gate.

4.4 Restriction set

After structural binding and applicability are established, the following restrictions are jointly required. Every restriction and structural invariant is nonexceptionable.

ID	Required permission predicate	Scientific role and false state
UQ-01	Resolved selection is exactly <code>SCI-PTC:uniform_constant@draft-0.1</code> , with no numerical parameter or override.	Prevents aliasing another constant or family. A known mismatch is ineligible; missing/conflicting selection is structurally unavailable.
UQ-02	Given an already established exact g, p, o and CAL-parent binding, the SCI-PTC compatibility record explicitly permits that coefficient generation for that product generation.	Prevents cross-generation borrowing after a valid question exists. Explicit incompatibility is ineligible; a missing or conflicting identity/binding is structural and decision unavailable.
UQ-03	$\mathcal{D}_{g,o}$ exactly equals the bound PTC output support, is finite and nonempty, and contains no duplicate or ambiguous identity.	A known empty population prevents realization; changed or conflicting population requires a new generation and makes the attempted binding unavailable.
UQ-04	The raw scalar is exactly dimensionless $c_\star = 1$ and the operator is exactly $\mathcal{N}_{\text{mean1}}$ on $\mathcal{D}_{g,o}$.	A different constant, unit, operator, or population is another family/version; known substitution is ineligible.
UQ-05	Within an established target binding, complete producer fidelity evidence establishes a published exact-one scalar and exactly-once stored membership for every and only target identity.	A complete authoritative violation is F , producer-invalid, and ineligible, including corruption even when unreadable and any missing, extra, or repeated stored member. Absent evidence or unexplained unreadability is U and decision unavailable absent another false. Structural failures stop earlier; unity is never substituted.
UQ-06	The requested named consumer has explicit permission in this exact family/version record.	Missing MAP permission does not affect JINC and vice versa. Authoritative absence is consumer-specific ineligibility; missing or conflicting permission is decision unavailable.
UQ-07	The coefficient generation and profile evaluation are immutable, one-way linked, and internally consistent; no earlier product or generation was overwritten.	Known lifecycle violation is ineligible; incomplete lineage is structurally unavailable.

PTC post-fit diagnostics, fitted loadings, sample validity, output retention, coordinate validity, MAP admission, JINC admission, and upstream response or covariance availability are not restrictions of this coefficient-fidelity profile. Their exact states and causes remain visible. Response and uncertainty availability have the owner-supplied role **advisory**: they contribute neutral permission and cannot determine coefficient eligibility. No identity response or zero uncertainty is substituted.

The realized `SCI-PTC:uniform_coefficient_handoff@draft-0.1` evaluation is the complete PTC coefficient-use QC record. No second circular producer-QC-passed predicate is required or permitted. UQ-05 supplies the independently

evaluated producer payload fact; the profile combines it with UQ-01–UQ-04 and UQ-06–UQ-07 under VAL’s knowledge algebra.

5 Proposed boundary/profile successors

5.1 MAP

The proposed boundary identity is `SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.2`. It changes only the coefficient facet needed for this family: a selected exact `SCI-PTC:uniform_constant@draft-0.1` payload may occupy the existing coefficient slot only after an approved family record explicitly grants SCI-MAP permission and the exact `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision is requested, applicable, eligible, and realized. The payload is dimensionless $\gamma_i = \omega_i = 1$ on the matching occurrence. Every frozen MAP identity, signal, retention, coordinate, response, uncertainty, exposure, influence, admission, placement, membership, contribution, failure, and classification rule remains unchanged.

MAP must verify exact family, version, generation, parent, occurrence, support, selection, profile, and permission binding, retrieve the authorized payload, and classify it independently. Only a finite strictly positive value may enter its existing arithmetic. The family does not make an occurrence MAP-admissible and defines no MAP-local factor. Until the family, profile, boundary, and exact bindings are adopted and active, the route is unavailable.

5.2 JINC

The proposed boundary identity is `SCI-PTC_TO_SCI-JINC_BOUNDARY v0.2-draft.1/r0.2`. It changes only the JINC-facing PTC coefficient facet: the same selected `SCI-PTC:uniform_constant@draft-0.1` payload may enter only after an approved family record separately grants SCI-JINC permission and the exact `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision is requested, applicable, eligible, and realized. JINC independently classifies $\omega_i = 1$ and, after all existing JINC gates, applies Equation 7. This creates no PTC ownership of κ_{ip} , signed normalization, conditioning, support, or bundle semantics and no base-v0.1 JINC response, covariance, or uncertainty product. All other frozen JINC constraints remain unchanged. Until separate JINC permission and every exact binding is active, the route is unavailable.

5.3 No inference or fallback

MAP permission never implies JINC permission and JINC permission never implies MAP permission. Neither consumer may infer this family from its unit, the numerical value one, another consumer’s availability, a label, or a missing selection. Absent selection without an authorized mode default, unregistered family, incompatible generation, missing payload/QC, or missing permission makes the route unavailable. There is no unity, alternate-family, neighboring-product, direct-CAL, no-op-PTC, reconstruction, or generation-repair fallback. PTC-disabled operation remains the RTC-terminal export route and creates no PTC, MAP, or JINC product.

6 NOI compatibility limit

This proposal leaves the selected NOI design law unchanged. On an exact MAP-admitted population, Equation 8 follows only because the proposed PTC coefficient is exact unity. G_{pi} , the admitted population, network stratification, MAP plan, and all NOI balance meaning retain their existing owners.

NOI may use the route only if its required parent record binds the exact MAP product/generation, G_{pi} , `SCI-PTC:uniform_constant@draft-0.1`, coefficient generation and payload, exact parent a_{pi} , and a canonical lossless rational decoding exactly to the MAP value. Unity itself has canonical rational form $1/1$; this does not prove that arbitrary G_{pi} or a_{pi} has a required lossless rational source. If any exact parent binding is absent, NOI design resolution remains unavailable. This proposal does not register a PTC NOI-balance family, authorize a JINC-hosted NOI route, change MAP arithmetic, or approve any of NOI’s four pending profile successors.

7 Numerical construction

7.1 Occurrence index and broadcast

Let p identify one exact realized PTC transformed-product/application generation for one observation o . Let g identify a new immutable coefficient generation bound to p , its immutable CAL parent, and the requested, effective-policy, observation-resolved, and realized family selection states. The logical coefficient index is the exact detector-time occurrence i . A compact representation may store one scalar only if it binds the lossless relation

$$\mathcal{B}_g(1) = \{(\iota_i, 1) : i \in \mathcal{D}_{g,o}\}. \quad (1)$$

Here $\mathcal{B}_g$ is the authoritative target binding between the identified scalar publication and the exact, unique population; it is established before payload fidelity is evaluated. A stored payload-membership manifest is separate evidence about whether the publication realizes that target. Time, row, column, shape, cardinality, detector label, coordinate equality, or container position cannot establish either the target binding or its stored membership. A missing or conflicting target binding, or a duplicate or ambiguous identity in the target population, prevents establishment of the question. After the target is established, a complete authoritative stored manifest that omits, adds, or repeats a member is a payload-fidelity failure.

7.2 Support and population

The proposed coefficient support and normalization population are identical:

$$\mathcal{D}_{g,o} = \mathcal{S}_{g,o}^{\text{coef}} = \mathcal{S}_{p,o}^{\text{PTC,out}}. \quad (2)$$

The family covers every and only exact occurrence in the PTC output support of its bound generation for one observation. The population spans all detectors, times, groups, segments, arrays, and networks represented in that support; none defines a separate normalization population. Observations and product generations are never pooled.

Equation 2 does not make sample validity a support predicate. An occurrence may carry coefficient one while its signal, coordinate, retention, or consumer admission is independently unusable. An occurrence outside the support has no coefficient under this family; absence is not zero.

7.3 Fixed constant and normalization

For finite nonempty $\mathcal{D}_{g,o}$, define

$$c_\star \equiv 1, \quad [c_\star] = 1, \quad (3)$$

$$\tilde{\omega}_i = c_\star, \quad i \in \mathcal{D}_{g,o}, \quad (4)$$

$$m_g = \frac{1}{|\mathcal{D}_{g,o}|} \sum_{j \in \mathcal{D}_{g,o}} \tilde{\omega}_j, \quad (5)$$

$$\omega_i = \mathcal{N}_{\text{mean1}}(\tilde{\omega}_i; \mathcal{D}_{g,o}) \equiv \frac{\tilde{\omega}_i}{m_g} = 1. \quad (6)$$

The unit 1 is dimensionless. Cardinality and summation operate on exact logical identities; a duplicate identity is a structural conflict rather than another population member.

The family has no configurable numerical parameter. A generic positive raw constant c would give

$$\frac{c}{|\mathcal{D}|^{-1} \sum_{j \in \mathcal{D}} c} = 1,$$

so its scale would be non-identifiable. The fixed value $c_\star = 1$ is therefore part of the family definition. A request carrying any override, including another positive constant, is not an instance of this family and cannot be silently ignored, clipped, or normalized into acceptance.

The explicit normalization declares a stable comparison convention. Changing the operator or population changes scientific identity even though this constant would remain numerically one. A singleton population is sufficient. An empty population has no mean and creates no generation or payload; it is not repaired by an empty pass, zero, or a scalar with inferred empty broadcast.

7.4 Consumer consequences

At the proposed JINC boundary,

$$w_{ip} = \kappa_{ip}\omega_i = \kappa_{ip}. \quad (7)$$

The equality concerns only the PTC factor and transfers no JINC ownership.

For the already selected NOI design on an exact MAP-admitted contribution population,

$$a_{pi} = G_{pi}\gamma_i = G_{pi}, \quad B_d = \sum_p \sum_{\substack{i \in C_p \\ \text{detector}(i)=d}} a_{pi}. \quad (8)$$

This does not define G_{pi} , change MAP arithmetic, or change the NOI design law.

8 Assumptions, decision semantics, and lifecycle

8.1 Scientific assumptions and limits

This definition assumes only that one exact realized PTC product generation exists for one observation, its exact output support can be identified, and the support is finite and nonempty. It assumes no statistical distribution, stationarity, independence, homoscedasticity, detector equivalence, exposure equivalence, residual model, sensitivity model, or response model. It neither estimates nor ranks data quality.

The coefficient value is an exact policy constant and therefore introduces no sampling uncertainty. That statement is limited to the coefficient itself. It does not assign zero to calibration, signal, response, covariance, cross-observation, support-selection, family-selection, representation, or performance uncertainty. Unknown remains unknown and absent covariance never means independence.

8.2 Knowledge and four independent axes

Every applicable fact and profile predicate uses $\{\mathbf{T}, \mathbf{F}, \mathbf{U}, \mathbf{C}\}$ with the admitted VAL meanings. Silence is $\mathbf{U}$. A structural contradiction blocks establishment of the question. A non-gating $\mathbf{C}$ is preserved and composes as unresolved $\mathbf{U}$. One decisive $\mathbf{F}$ yields ineligible even when another non-gating predicate is unresolved; without a false, any required unresolved predicate yields decision unavailable; all required predicates true yields eligible. Every cause is retained without evidence-erasing short circuit.

The `SCI-PTC:uniform_coefficient_handoff@draft-0.1` record carries independent axes:

- R = requested or not requested. A family not selected and handoff not requested has no eligibility evaluation.
- A = applicable, inapplicable, or applicability unknown under the profile definition. Known outside-support occurrences have $E = \emptyset_E$.
- E = eligible, ineligible, decision unavailable, or uninstantiated $\emptyset_E$ where no question is evaluated.
- Z = realized, incomplete, failed, or not produced for the profile evaluation. Coefficient payload availability remains a separate typed fact.

A realized decision-unavailable record is a successful account of missing authority. If the evaluation artifact itself is incomplete, failed, or not produced, it cannot manufacture an eligibility assertion.

A complete authoritative UQ-05 false is likewise a successfully realized negative profile evaluation: Z = realized, E = ineligible, and the separate typed payload state is invalid. It does not set profile-artifact Z to failed. That failure state is reserved for failure to complete or produce the profile evaluation artifact, independent of whether a completed predicate is true, false, unknown, or conflicting.

8.3 Payload order and failure scope

Processing first establishes authoritative identity, parentage, profile authority, applicability, the exact unique population, and the target broadcast binding. Only then does it evaluate the bound payload and its stored membership for producer fidelity. A decisive fidelity false excludes the payload; an unknown or conflicting fidelity fact remains decision unavailable

absent another false. Finiteness or a numerical value of one cannot supply missing authority, validity, support, membership, or permission. MAP and JINC subsequently perform their own positive/zero/negative/non-finite/ unrepresentable classification.

Condition	Required result	Failure scope
Missing/conflicting Registry, source, parent, occurrence, generation, support, payload-publication identity, or target broadcast binding; or duplicate or ambiguous identity in the authoritative population	Applicability unknown or decision unavailable; UQ-05 is not evaluated and there is no fallback payload	Entire affected coefficient generation
Known empty $\mathcal{D}_{g,o}$	No family instance or payload; no eligibility claim	Observation/generation
After structural identity is established, explicit incompatible-generation permission, wrong selected family, or known family-definition operator/unit/population substitution	Ineligible with all causes; no reinterpretation	Entire scalar-broadcast generation
Complete authoritative payload-fidelity check establishes decoded non-one or wrong unit, digest/integrity failure, corruption, truncation, decoder-invalid bytes, known unrepresentability, or missing/extra/repeated stored membership relative to the established target	UQ-05 is F ; the payload is producer-invalid and the realized profile decision is ineligible, even when corruption makes the scalar unreadable	Entire scalar-broadcast generation
Payload, bytes, decoder, digest, integrity result, complete check, or complete stored membership manifest is absent/unavailable; or payload is unreadable without an authoritative fidelity cause or verdict	UQ-05 is U ; typed payload state remains unknown/unavailable and a realized profile decision is decision unavailable absent another false	Entire scalar-broadcast generation
Absent explicit consumer permission	Ineligible for that named consumer only	Consumer-specific route
Occurrence authoritatively outside support	Inapplicable, $E = \emptyset_E$, and no coefficient	Occurrence
Consumer numerical rejection of an otherwise faithful payload	Preserve PTC record; consumer excludes or fails under its rule	Occurrence and consumer
Unrelated metadata/advisory diagnostic missing or conflicting	Preserve state and cause; neutral for this profile	No coefficient-QC failure

8.4 Immutable lifecycle and repetition

Creating g never changes transformed product p or an earlier coefficient generation. A change in support, product/application generation, observation, family/version, operator, population, unit, profile, permission, or selection lineage creates a new scientific identity and preserves all earlier parents.

Repeated realization from the same exact inputs must yield the same mathematical payload and membership relation, but each realization retains its own immutable generation identity. Equal values do not authorize substitution or borrowing. Cross-generation use requires an explicit compatibility record binding both generations and preserving their claims. This proposal provides no general compatibility alias.

9 Scientific validity and uncertainty status

Claim or uncertainty	Status class	Meaning in this draft
Frozen ownership and generic coefficient constraints	Approved/recovered	Adopted from the admitted packet and not re-opened.
Uniform definition, normalization, support, and profile	Unresolved decision	Complete author proposal awaiting owner approval.
Separate MAP and JINC permissions	Unresolved decisions	Proposed for both; each remains independently open.
Algebraic positivity and mean-one identity	Engineering conformance claim	Follows analytically for exact one on a finite nonempty population; eventual artifact tests must confirm exact realization.
Coefficient sampling uncertainty	Known by definition	None is introduced by the fixed constant; this is not a zero estimate of physical uncertainty.
Family- and support-selection uncertainty	Known but not supplied / unresolved	Not encoded in the coefficient. Unknown is not zero.
Signal, calibration, response, covariance, and cross-observation uncertainty	Producer facts, possibly unavailable	Preserved without reinterpretation; absence does not imply zero or independence.
Implementation conformity	Implementation unassessed	No implementation was inspected and no conformance claim is made.
Representation and response fidelity	Validation not performed	Exact encoding, broadcast, lineage, and whole-chain behavior require later evidence.
Observational performance and comparative suitability	Validation not performed	No sensitivity, variance, bias, or optimality claim is made.
Production readiness	Validation not performed	No route is authorized by this draft.
NOI profile successors	Deferred/unavailable	All four retain their admitted proposed, unapproved, Registry-unbound, and unevaluable status.

10 Decision boundary

The draft has no blocked mathematical definition. Every consequential new choice remains an open author proposal in the owner-decision ledger. Until those choices are adopted and exact successor source/Registry bindings exist, the family, profile, permissions, and boundary successors remain unavailable for evaluation or use. Approval of this family would not close deferred families, mode defaults, or independently governed route prerequisites.

11 Compact owner-decision register

This common-source register is explicitly checked against the companion `OWNER_DECISION_LEDGER.md`; the ledger carries the full owner, evidence, resolution-authority, date, and affected-document fields.

Recovered and deferred states

IDs	State	Disposition
UFC-REC-001-007	decided	Admitted PTC, registry, MAP, JINC, NOI, VAL, and PTC-disabled-route facts are recovered constraints; no claim is blocked.
UFC-DEF-001-005	deferred	Other coefficient families, defaults, NOI profile successors, performance criteria, and a general cross-generation alias remain outside this draft.

Open author proposals

ID	State	Exact blocked claim or output
UFC-OD-001	open	Registered exact-unity numerical family and payload.

ID	State	Exact blocked claim or output
UFC-OD-002	open	Exact logical index and scalar-broadcast declaration.
UFC-OD-003	open	Approved support and normalization population.
UFC-OD-004	open	Approved mean-one operator and empty/singleton behavior.
UFC-OD-005	open	Approved no-parameter selection rule.
UFC-OD-006	open	Complete evaluable coefficient-handoff profile identity, use, object, and applicability.
UFC-OD-007	open	Approved UQ-01–UQ-07 truth rules, disjoint structural and payload-fidelity cases, no-exception rule, advisory roles, and failure scopes.
UFC-OD-008	open	SCI-MAP permission and MAP use of this family.
UFC-OD-009	open	SCI-JINC permission and JINC use of this family.
UFC-OD-010	open	Immutable MAP and JINC successor-boundary bindings.
UFC-OD-011	open	Exact actual-owner field for active Registry/profile binding; the mathematical draft is not blocked.
UFC-OD-012	open	Approved generation/replay and cross-generation rules.
UFC-OD-013	open	Approved coefficient uncertainty meaning.
UFC-OD-014	open	Active coefficient Registry identity and source binding.

Precise open question

ID	State	Exact question and blocked output
UFC-Q-001	open	Is the actual scientific owner Grant Wilson acting for SCI-PTC, as proposed, or another named authority? This blocks active Registry/profile binding only.

12 Normative proposed requirements

Every requirement below is a proposal pending scientific-owner adoption.

ID	Requirement
UFC-REQ-001	The family shall have exact identity <code>SCI-PTC:uniform_constant@draft-0.1</code> , scientific owner SCI-PTC, and no active or registered status until owner approval and exact Registry binding.
UFC-REQ-002	The logical index shall be exact detector-time occurrence i ; any scalar publication shall structurally bind its identity to the complete lossless target relation in Equation 1. Its stored membership manifest shall be separately evaluated as payload-fidelity evidence.
UFC-REQ-003	For one observation and bound PTC product generation, coefficient support and normalization population shall both equal the exact PTC output support as in Equation 2.
UFC-REQ-004	On a finite nonempty population, the raw constant shall be exact dimensionless one and normalization shall be the arithmetic mean-one operator in Equations 3–6.
UFC-REQ-005	The family shall accept no numerical parameter. Any constant, unit, operator, or population override shall require another registered family or version and shall not be silently ignored.
UFC-REQ-006	A singleton population shall be valid. An empty population, or an authoritative population with duplicate, ambiguous, or conflicting identity, shall create no realized coefficient generation and shall not be repaired by zero, unity, deduplication, or an empty pass. Stored payload membership is assessed only after a unique target population exists.
UFC-REQ-007	Occurrences outside coefficient support shall have no coefficient under this family; absence shall remain distinct from numerical zero.
UFC-REQ-008	Every generation shall bind exact transformed product, application generation, CAL parent, observation, selection lineage, support, family, profile, and publication identities without mutating any parent or earlier generation.
UFC-REQ-009	Repeated realizations shall reproduce exact values and broadcast membership but retain distinct immutable generation identities; numerical equality alone shall not permit borrowing.
UFC-REQ-010	The family record shall state SCI-MAP and SCI-JINC permissions separately. Each proposed permission requires independent owner adoption and neither shall imply the other.

ID	Requirement
UFC-REQ-011	Coefficient fidelity shall be evaluated only under the complete proposed <code>SCI-PTC:uniform_coefficient_handoff@draft-0.1</code> record; the unsupported generic coefficient-QC placeholder shall not be used or simulated.
UFC-REQ-012	The profile shall bind exact use, owner, source, object/domain, restrictions, no-exception rule, response/uncertainty roles, compatibility, supersession, missing/conflict behavior, aggregation status, and lifecycle.
UFC-REQ-013	Structural target identity, stored payload membership, payload availability, producer fidelity, sample validity, MAP admission, JINC admission, and each consumer's numerical classification and contribution shall remain distinct.
UFC-REQ-014	Every applicable fact and predicate shall use T/F/U/C semantics; silence shall be unknown, conflict shall be preserved, known exclusion shall dominate unrelated unknowns, and no evidence shall be erased.
UFC-REQ-015	Requested, applicability, eligibility, and realization shall be four independent axes, including uninstantiated eligibility where no question is evaluated. A completed false predicate shall yield a realized ineligible profile evaluation, not a failed profile artifact.
UFC-REQ-016	All profile restrictions and structural invariants shall be nonexceptionable. No exception from VAL, another use, consumer, or mode shall alter them.
UFC-REQ-017	Structural missing/conflict/ambiguity shall stop the question; within an established target, absent evidence or unexplained unreadability shall be U , while authoritatively established corruption or missing/extra/repeated stored membership shall be F . These states, conflict, exact zero, non-finite, and unrepresentable cases shall remain distinct and follow the stated failure scopes.
UFC-REQ-018	Response and uncertainty availability shall be advisory for this coefficient-fidelity profile; exact upstream states shall be preserved and neither identity response nor zero uncertainty shall be inferred.
UFC-REQ-019	The coefficient shall be nonprecision and shall not be interpreted as inverse variance, sensitivity, significance, fitted loading, scatter, validity, exposure, response, covariance, independence, or optimality.
UFC-REQ-020	The MAP successor shall change only its coefficient binding, retain all MAP-owned arithmetic and gates, and require MAP's independent finite-strictly-positive classification before use.
UFC-REQ-021	The JINC successor shall change only its PTC-coefficient binding, retain JINC ownership of κ_{ip} and its arithmetic/gates, and create no new JINC response, covariance, uncertainty, or bundle role.
UFC-REQ-022	NOI compatibility shall require the frozen exact lossless rational-parent record for the exact MAP and coefficient generations. Missing binding shall leave NOI unavailable and shall not change its design law.
UFC-REQ-023	No family or mode default shall be inferred or proposed. Missing selection or authority shall make the route unavailable without fallback.
UFC-REQ-024	Every material change of definition, support, operator, population, unit, profile, permission, boundary, or missing/conflict behavior shall create a versioned successor and shall not rewrite an earlier evaluation.
UFC-REQ-025	Algebraic conformance, implementation conformity, representation/response fidelity, observational performance, and production readiness shall remain separate evidence claims.

13 Predictions and discriminating edge cases

These predictions follow from the proposal; they do not claim that a system has been tested.

ID	Prediction or edge case
UFC-PRED-001	For every finite nonempty exact population and every member, $\tilde{\omega}_i = \omega_i = 1$ exactly; every pairwise coefficient ratio and the arithmetic mean are one.
UFC-PRED-002	A population with exactly one occurrence is valid and produces one exact coefficient.
UFC-PRED-003	An empty population produces no mean, scalar broadcast, or realized family payload.
UFC-PRED-004	Splitting, merging, relabeling, adding, or removing groups, segments, arrays, networks, detectors, or times leaves member values at one when support is otherwise valid, but any support change creates a new generation identity.

ID	Prediction or edge case
UFC-PRED-005	Repeated realizations from the same exact inputs are numerically identical, yet substituting one generation for another without an explicit compatibility record is forbidden.
UFC-PRED-006	A request containing $c \neq 1$, even when $c > 0$ and mean normalization would cancel it, does not select this family and cannot be accepted by silently dropping the parameter.
UFC-PRED-007	After structural establishment, a complete authoritative check that identifies a decoded non-one or wrong unit, corruption even when unreadable, known unrepresentability, or missing/extra/repeated stored member makes UQ-05 false, the payload producer-invalid, and the profile ineligible. The negative profile evaluation is successfully realized rather than a failed profile artifact. Exact zero, negative finite, or non-finite content also fails the consumer's independent positive classification.
UFC-PRED-008	An absent payload or fidelity input, or unreadability without an authoritative fidelity cause or verdict, makes UQ-05 unknown and the profile decision unavailable absent another false. Unity cannot be substituted.
UFC-PRED-009	Missing/conflicting target broadcast binding or a duplicate/ambiguous identity in the authoritative population prevents structural establishment and yields decision unavailable. A stored scalar one cannot repair it.
UFC-PRED-010	An invalid signal occurrence can still have a faithful coefficient one. It contributes only if all independent producer and consumer gates admit the signal and occurrence.
UFC-PRED-011	MAP consumption gives $\gamma_i = 1$ and removes only relative variation attributable to the PTC coefficient; it does not make total MAP contributions equal.
UFC-PRED-012	JINC consumption gives $w_{ip} = \kappa_{ip}$ for the PTC factor, including the sign supplied by JINC; it does not make JINC weights positive or uniform.
UFC-PRED-013	MAP permission alone leaves the JINC route unavailable, and JINC permission alone leaves the MAP route unavailable.
UFC-PRED-014	With no selected family and no authorized mode default, neither consumer may infer this family from a stored one, a label, or another route.
UFC-PRED-015	On an otherwise valid exact MAP host route, $a_{pi} = G_{pi}$; missing lossless rational identity for that exact a_{pi} still leaves NOI design resolution unavailable.
UFC-PRED-016	PTC-disabled or missing positive-rank PTC realization produces no instance of this coefficient and creates no MAP or JINC route.

14 Evidence required after owner adoption

Eventual conformance evidence must bind the exact approved source and Registry records. Algebraic tests must cover nonempty populations across detector, time, group, segment, array, and network labels; singleton support; exact mean-one results; and exact scalar broadcast membership. Negative tests must cover empty, duplicate, ambiguous, or changed authoritative support; parameter overrides; missing/conflicting target broadcast binding; wrong unit/operator/family; absent payload or fidelity input; unexplained unreadability; authoritatively established corruption or unrepresentability; missing/extra/repeated stored membership; non-one content; wrong product or generation; absent consumer permission; and cross-generation borrowing. Replay evidence must show exact numerical reproducibility while retaining distinct immutable generations.

Profile evidence must exercise T/F/U/C composition, all four decision axes, nonexceptionability, structural versus non-gating conflict, payload ordering, cause preservation, and every failure scope. Boundary evidence must show MAP and JINC performing independent numerical classification and admission, permissions not crossing, and no fallback route. NOI evidence, if that independently governed route becomes evaluable, must bind the exact MAP value and canonical lossless rational parent without changing MAP arithmetic.

Scientific validation is a separate future activity. It requires owner-approved observational or simulation populations, comparison questions, metrics, uncertainty treatment, and acceptance criteria. None is invented here. Such evidence could establish comparative performance or suitability; algebraic tests cannot. Production readiness adds operational completeness and failure-handling evidence only after scientific and implementation conformance are established.

A Engineering interpretation of the shared core

An eventual implementation assessment should treat UFC-REQ-001–010 as the family payload and lifecycle surface, UFC-REQ-011–018 as the PTC-owned profile/evaluation surface, UFC-REQ-019–024 as consumer, NOI, fallback, and change-control boundaries, and UFC-REQ-025 as the evidence ceiling. This grouping changes no requirement meaning. The authoritative mapping for review is the companion **CROSSWALK.md**; exact draft identity and provenance bindings are in **PROPOSED_REGISTRY_AND_BOUNDARY_RECORDS.md** and the artifact manifest.

Conformance evidence must report exact source, family, profile, product, coefficient-generation, observation, occurrence-support, consumer, and evaluation identities. A numerical array of ones by itself establishes none of those bindings.